

ActInf Textbook Group ~ Cohort 7 ~ Session 25

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All right. Hello fellows. We are in by all independent accounts the last of the meetings of cohort 7 in chapter 10. We've already had some interesting sharings and consolidations. So what would be interesting in this time? How how wh where if chapter 10 is the takeoff, where do we go? Or where are we in our own chapter 10? Well, for me, for me, you say Stephen. Yeah. I was just going to say it's model building. It's applying applications is where I'm at. Andrew. Um well, for me, um it's just an achievement to have finished reading the book. I still have to read the appendices, but um uh upon reading the book, I feel like I have a sense of the complete picture, maybe in a murky way. Uh and then what it invites me is a couple of u things. One is to be able to tease out the flesh and the bones I guess or or to say like what is actually active inference saying and then what is kind of like additional ideas that come out from applying active inference because those additional ideas they could be brilliant or they could be just not so brilliant. Uh but that's a separate thing. So um and you were just talking about that if you just look at the bare bones active inference that's actually a common theme across all different types of let's say um I think it was like learning you know large language models or or different types of AI approaches. Um so um what that means is that in order to really appreciate active inference it's important to look at it minimalistically like not what are all people trying to do to apply but really what is it like? It's a little bit for me like calculus like calculus is absolutely important for modern physics. Um and so you wouldn't want to make calculus more physical than it is. It's just a mathematical um way of thinking and then you wouldn't maybe want to say that the calculus is a theory of everything but to be able to really appreciate so what exactly is calculus and what is it contributing you know and how does that relate to linearity let's say or not uh and things like that uh I think that's where I'm at and I think like in order to do that maybe the next step for me is just to uh compute you know so uh to learn how to uh use the Python uh modules I guess and to compute things that would be relevant for me and get some hands-on experience. I think that's probably the next step for me.

And then maybe after that like then to say okay take a fresh look at let's say what are Bayesian statistics and the related math and stuff and work from there. That's where I'm at. And maybe to say like uh what's attractive for me? It does seem uh to be um very sweet in terms of putting perception and action in a unified footing. But that's a very particular case where you're assuming that you're looking at the world with a generative model and then from the point of view of that model you know you have this active look at perception and so you have this active look at action let's say and then you have this ability to quickly switch from one to the other based on I guess free energy um the free energy uh I guess maximizing or minimizing the free energy So that's all fine, but that's assuming a certain, you know, outlook. Uh, and I don't think that every outlook is always based on a generative model. Um, so there's alternative outlooks, let's say. But, um, so just to kind of tease that out, what what kind of alternative outlooks other than looking at systems in terms of like their input and their output? Well, I just don't think like I don't think necessarily that the physical world is based on a generative model, you know, because a generative model is kind of like a it's a symbolic assumption that like so in the way I would think about like um there's knowing versus not knowing, you know, so like there's you know maybe values and variables and so um the generative model presumes that you have this world of variables of not knowing of symbolic language um But I think that something like a neural network um doesn't have that. I think it's it's not about u having variables and slots. I don't think that that's a good way to uh think of a neural network uh working or or just all different kinds of wiring working. Um there's there's a and so I think more like a generative model is working on top of other things but not it wouldn't make sense I think to have a like so the notion that everything is a Markov blanket. I just don't think that that makes sense. You can look at everything as if it's a Markov blanket, but that's very peculiar that's that's a peculiar maybe important way of looking at things, but that doesn't uh that's not the only way that things would function. And I don't think physics function. Well, I mean it returns to kind of map territory there because when you said that the neural network isn't just parameters and slots, I mean on the first level, it literally is. That is what the parameters of the neural network is. They literally are parameters and slots. And then even abstracting if we couldn't look at those parameters in the slots, there's still um an input output relationship which is modeled with an interface. Mhm. That's kind of the the the surface like the sort of cloud of unknowing edge of our um doors of perception that any modeler finds themselves in which is the situation described by inferring latent states from observations and then the special case of action is not acting. So the passive case we get for

free and that's why we seek to study the possible joint um co-action of perception and motor movement for example. Well, I think maybe there's a two ways. See, I think it's more about I think your points are important and and I guess they're probably correct. U but I think it's maybe then about interpretation like even with a neural network and I very know very little bit little bit about them but I think that there's two different ways to interpret them. like you can interpret them in terms of like a hard wiring where like you're just getting signals that are coming together and they're you know at any inst at any instance there's no notion of variable I mean it's just what it is it gives you what it gives you now then you can say okay but let's make it and maybe that's the point of the neuron nature is that let's make it more flexible where it could be rewired right it doesn't have to be just wired once right it could be rewired so the notion that this is plastic and that this is rewirable and that these are slots, let's say, right? Like that's almost like a different interpretation of what's going on. It's saying, well, there's a different way of looking at the neural network where it's saying this is all plastic. These are all arbitrary. You see, and I think the fact is that those both those interpretations are important. And I my claim would be I think that both those interpretations are communicating with each other and there's a gap between those two communications. And so so that's the gap in the action perception model like the action the I mean the I'm sorry yeah well the organism and the environment in the environment you have one hardwired reality and then in the organism you have this idea that oh we can be flexible you know like we can be flexible but whereas the environment is considered it is what it is so I think that Um I I I think I see you on that view. Um and we sometimes are thinking about that like uh it's a static room and the agentic organism can navigate do the change. But the whole point is it's joint system modeling and it could be a super dynamic room with a metronome in it or it could just be two metronomes or it could be two really dynamic things. It's like so kind of making a qualitative assumption that that the environment or the organism has any given structure or that that one is larger. It it they're just convenient. It's like here and there like there could be here and and here could be there but well or which is the context and which is the being you know I think like that's the kind of question and so the thing I'm researching and you know a bit about this I call it you know it's predis it turns out that you have this oscillation between well who's the being and who's the context and so in quantum symmetries it's kind of you talk about the context of the context which could be the being, you know, so it's kind of like a you have a vector and you have a what's called the dual space. These are like functionals on the vector space like but then the dual of the dual is typically the vector where like you just it's the values I guess it could

take. So, and it turns out with a quantum symmetry when you do the dual of the dual or you do the context of the context, you may get back to where you started, but you're also allowed to get back to the negative of where you started, you know? So, it's kind of like the the in this case it'd be like the data became the slots and the slots became the data or like the particles became the holes and the holes became the particles. And so and the idea is that there's this thing called B periodicity where you can cycle through these things and there's like eight different possibilities and you just get back. So the thing I realized yesterday was when you go from context and back um and you look at this whole cycle, it looks like the database update problem where like you have two copies. We're kind of like looking at things from two copies and then you want to be able them to be mismatched. Like if you want to you're working with a database and it's giving you a presentation, you want to be able to edit that, but then it needs to be able to update its record. So there's going to be a mismatch and then there's going to be a re synchronization and a mismatch and recynchronization. So there's an oscillation between are your two different ways of looking at them kind of meshed so nicely together they're it's hard to distinguish or are you realizing no here's an instance and here's a general plastic, you know, ability. let's say and they need to be kept separate and they actually mismatch right now but then they'll rematch. Am I kind of making sense? It seems like that it would be again another a feature of the model that would employ the methods. Yeah. So this is saying like active inference is a very nice beautiful or maybe even slick like way of talking about one aspect of this and then you have to okay so what's what's the models that you're going to think of this in terms of and so and that's what I'm talking about and and I think the idea is that uh between the organism and the environment is there a third mind that is kind of like facilitating that or not you know is it needed or not but if you have this kind of like database update problem and or you have this kind of like mismatch like or like you know I have this emotional situation where like I don't feel things are right let's say right and you're trying to model that it may suggest that there's above and beyond that these things are not self-regulating there's a regulator on the side that's contributing to you know cybernetically managing it that's what like full-fledged consciousness would be okay this now is running up against something that I'm constantly struggling with. uh this um free energy principle is a natural uh uh uh feature of existence I feel and it's being this this framework um is being applied to our framework which our our human conceptual framework which is completely ruled by our conceptualizations I mean nothing we do lives outside of the constraint of how our minds work. Uh we think the world works the way it does because of the nature of our

construction, our bodily ability to conceptualize and these concepts are constrained by the way we're constructed. uh and I run up against um uh working back and forth between these two realities. The the world is the way it is and works the way it does and we interpret it the way we do uh to the best of our ability. Um there this uh I I I posted a reference to the handbook of abduction. Uh you remember Daniel the \$700 book that Oh yes. worth worth every penny which which kind of u confronts this problem of it's basically an epistemological problem. How do we know what we know and what constrains what we can possibly know? But again, everything I'm interested in with active inference is through the filter of how the brain works, how the mind works. And um the I was just at a seminar yesterday on is it possible for machine intelligence to have meta awareness and that's something that that that this lab is working on right now. Uh the PI is going to a conference in San Diego of um I don't know what it's called but it's machine learning. It's some big leagues trying to address this problem of how can these deep learning models be aware of themselves to the point where they can uh filter out hallucinations themselves where they can know or assess how good their reasoning is. Um so again on that in that vein and I thought of you when I read this paper that I posted in the chat um modeling meta awareness because uh I know you were working on this three minds three brains issue and it explicitly addresses that uh in a way talking about hierarchical model of meta awareness and attention control, perception, attentional states, and then meta awareness states, which I think is kind of like what you were describing. That's right. Okay, that's how I'm glad I I copied the link and I'm glad that you this. Yeah, it's funny that that that had come up in some other who knows when chapter 10 discussion, but I added it in those kind of it was kind of a Hallmark paper certainly in terms of connecting multi-scale modeling. And then here's one from that I um put in the notes too from a few with um Lars Sammit Smith, first author in that 2021 paper and right in others. I mean, this this gives some some very interesting angles. Um, and they do test out some things with LLM, but like when we kind of get against that doors of perception, like beyond that classic Blake quote, I think of this this letter. Yes. As a man is, so he sees. And it's like, okay, well then, you know, pack it up. Nothing to see here. It's like know everything to see here, but then how to balance like the world as it is, doors of perception being cleansed, seeing the world, you know, infinity in the flower and all of that with the solipsism of anthropocentrism. It's like, well, how could we escape anthrocentrism when we're our own egocentric navigational unit and we're seeing things through our own perception and cognition apparatus. So, it's like what could be the escape from anthropocentrism? And then this part um you certainly mistake when you say that the

visions of fancy are not to be found in this world. Like I see that as the reality of cognitive science, speculation, counterfactuals, like expected free energy rollouts, those are in the world. That doesn't mean that they're justified true belief. Doesn't mean that that the um parameters as they're estimated to be would be any level of accuracy. But visions of fancy are found in the world and and that goes in so many different ways. I mean taking us all the way up into sort of matter first ontologies which lead us to the hard question of consciousness or mind first ontologies that that have a uh counterveiling narrative on the emergence of stability. Uh so I have a I have a question uh of something whether it could be modeled naturally with active inference or if it's something different you know which relates to this and so because I think of active inference as kind of like the second mind type of issue uh but the first mind would be just direct experience whatever that means like the answers that are coming out unconsciously popping into my brain let's say whatever and uh so there's these things I talk about as doubts and counter question. So, um, once there I was given this question, well, how do you know you're not a robot, right? And so, the it's a difficult question because it puts everything in doubt. Like, you know, if I'm a robot, how can I trust anything I know? And so, I think that there's really only one way out of that um aside from just falling back to sleep. But, u if you really want to take that question seriously, you ask a counter question. You go, well, would it make any difference? Okay. Right. And and the the way the counter question works, it says, look, I'm not going to rely on anything I know. So the ability just kind of like clean the slate to say, look, and then it's kind of like you tell me what a robot is, right? Like maybe I am a robot, you know, or maybe I'm not a robot, but you give me some criteria so that I could tell like, you know, do robots bleed or do they not bleed or, you know, do robots have feelings? Can robots fall in love? You know, what is it? And then maybe I'm a robot, but maybe I'm not. But I'm not going to rely on my I mean I'm not going to I'm just going to let go of my experience and build it back up. So what that does from the first mind's point of view, it's a way of letting go of itself and saying there's something maybe more out there like to get out of your context. And so the things I was talking about like how do you go from context to context to context? This is the way of letting go of your current context and being able to get into you know without a context and into another context let's say. So I think and so there's I I I did like 30 examples and there was like it ended up being like seven patterns, you know, of counter questions. So like is this do I truly like this? Well, how does it seem to me? Do I truly need this? What else should I be doing? Is this truly real? Would it make any difference? Is this truly problematic? What do I have control over? Uh is this truly reasonable? Like am I able to consider the

question uh is this uh truly wrong? Uh is this the way things should be? um am I truly anxious? Well, what am I doing this about this? And then there's an eighth one that just whenever you want, you can say, well, what do I truly want? You see? So, so there's this whole apparatus of these things. And I'm just wondering, I think this is like before you even get to having a generative model. This is how you start to become letting go of yourself and starting to operate with a generative model that's abstract, that doesn't know anything, saying, "I don't know anything, but what could I do with that?" This is like the preliminary in order to build up a but or to to to get back to thinking in terms of a generative model. But where would this this type of thinking fit with active inference? I don't know. Oh, I think it's there's there's no goal. I think the it's it doesn't have a goal, right? like what do I truly just that's completely compatible with the the the kind of model that uses active inference there are no priors and then it all your prior with experience well so what if you don't have priors then what happens with active inference Daniel you can answer that it don't sort there's the like sort of uh substrate fabric of reality question like what is what is the sound of the silent bell like a like if there really were no prior what would there be? So that that that sort of fades into more more axiomatical metaphysics. But let's just start with a case where there there there's a given prior but of a certain kind but it's a uniform within whatever kind it is. Mhm. and then popping up a level of context is like kind of structure learning like is that prior modifiable? Well, I think what this is maybe I didn't quite absorb what you're saying, but I think like what this is saying is like I'm going to let go of all my priors. Why don't you give me some priors? Right? like you know but see what what happens there is that instead of saying I have natural priors you know I'm wired in with certain priors like I'm saying my priors could be variable like let's go with any priors you see and that's a huge shift all of a sudden because you're in a different realm you're in moving from the first mind where there's these things that are hardwired or like prior that are hardwired to saying oh no priors could be variable you see yeah this is kind of addressed in the whole Buddhist tradition of of developing a along the spectrum of you know directed attention uh awareness to uh open attention open awareness to nonduality where there are no subject object differences which is kind of like no priors in a way. it that's like uh either there is you know there only either one awareness or no separation between awarenesses th this is sort of where this paper takes it with translating yes sorts of meditational prompts into um natural language and using that um in terms making parametric modifications like modifying the temperature value um like in a sort of uh infinite temperature scenario like just just you know the ball is vibrating on the landscape in the absolute zero context it's going to find a local

minima or maybe even the global minima. Um, in the infinite temperature setting, the landscape is functionally flattened. So, whatever structure the landscape had, as the temperature goes to infinity, it doesn't really matter what the temperature was. So, there's there's those kinds of parametric manipulations. And then there's like usage of natural as I mean delivering the payload itself within a given parametric configuration of the model. But I mean like this is what are you showing? What what paper is this? This is the um I'll I'll I'll put it in the in the Okay. Zoom chat as well. Like this is just from a few days ago. I mean um obviously with with Lars with with uh Chandaria. Yeah. Yeah. Yeah. They're they're they're kind of going there and uh I think it's inevitable. I think that all roads lead there. Uh you look at the development of for example um learning algorithms um machine learning algorithms uh or you know there are machine learning uh architectures of the uh cognitive architectures uh different ways of approaching how to model cognition. lots of different models and over the years they basically all have been converging into the same general architecture as as they modify themselves. And as I say, there's probably an inevitable convergence of all these different approaches because it's all trying to approach the same thing, the same truth, I guess, is what I would say. Yeah, I mean there there's there there's so many aspects on this and uh like I think of mind and super mind orbindo and many different wisdom cognitive methods and then it's like and and also there's another you know from another vision or another perspective It's purely quantitative structured formal accounting system for for cognitive things. Yeah. It's like it's that's the deflationary view is sort of it's just statistics. Now Blake might call that Newton's single vision with like the sort of sleep of technical expertise. Um, and and I mean it's it's embodied within our own ability to to see it those multiple ways or oscillate or or hold both open. Certainly, it's reflected in our communities where some people at at the syntactic or the semantic level are going to have allergies to to these kinds of prompts as well as allergies to mathematical symbols. So then it's like so then that and then that goes to kind of like Krishna Murdy like truth is a pathless land like but how could there be a place with no paths within or out? Yeah I I put in a link in the chat to a presentation I gave u about uh how we use variables in the mind. Uh and so like yeah if if you look there like at that any any of those like but you get the kind of idea that um when we look at a variable um there's so much hidden there because uh it's subject to interpretation like is it constant is it variable you know what does it mean for it to be fixed right like um is it just a label uh there's so much loaded in there um that if you try to tease it out you get all these you know I think like 24 different possibilities or something like that you know there's this huge uh risk richness that's being

hidden, let's say. So this idea like with the priors, are those prior something that are fixed and constant and just given, you know, personal or are they become all of a sudden they become variables that you can play with, you know, that someone else could be filling in for you, right? Like, and how do you what does it mean to switch back and forth between those types of um ways of thinking? Uh that's I think it's kind of there's this chasm that's kind of being glossed over. Um wow we really come down to the topic of communication the how can the conceptualization yes but then communication of those concepts and I this is why we always turn to rigorous mathematical you know symbology because it's the least vague, the least uh imprecise and yet we still have to know when what are the appropriate symbols to use. We can we can inappropriately apply statistical methods and you know uh obscure the communication even still and yet it's I think the best we have or there is the extended mind idea that I can have my conceptualization communicate it and because you're so similar to me in construction. You can create your own conceptualization that is more or less similar to what mine is based on the means of communication we're employing. And because we live in a structured environment that's the same, it constrains our conceptualizations in a way that are most likely going to be most similar. Yeah. So many ways to take this just just trying to just trying to triangulate it with like the abductive work that you've shared and and what they're what they're discussing in this paper like we're not imposing any single metaphysical system. However, what if there's a single meta metaphysical system? Huh. Yeah. And it's like another like the the sort of show not tell realization of metacognitive science is like narrative and perspectival plurality as popping up a layer in the consideration of of all of what had been considered at a at a cognitive layer. Yes, I think this is approaching the best we can do. So in trying to speak that language but um and I have I think you're all a little familiar with this what I call these three minds like uh so I would think like as a generative model as a type of metaphysics let's say or conceptual system and then the point being for a for a conceptual system to really function well and do its job there needs to be a tiny uh and very uh very clear it's it well there just seems to be certain things that that explain that it's or that maybe um amount to the fact that it's functioning, it's able to function. And those are the things that's able to be able to swap back and forth with the things it's trying to model. Like you know that it's if it's if it's a good model, you can invert it and you can the context can be understood as the as the reality and the reality the context and flip back and forth and all these symmetries. And there's very few symmetries you have to worry about. Like there's a there's but there's so many there's like eight. And if you can do that, you have a metaphysics of the metaphysics. And if you can't do

that, you simply don't have a function. You know, you have a you have a generative model that's just not going to function um very well. It's just going to fall apart. So um so the fact that the generative model fits up with the things it's modeling is what the third mind is all about and the symmetries that need to be satisfied for that to happen. There's which are like updating symmetries like the way a database would function. you know, how do you how does it update itself? So interesting. I I I I totally agree with what you're saying. Uh I guess and I um it it as a side tangent, I'm always amazed at why distillations of truths always come down to three parts. And I think that's got to be a natural principle. Well, sorry, it's not a natural principle of the world. It's a natural principle of our conceptualizing mind. There's this, there's that, and then there's the thing that's neither this nor that. You know, that's three parts. Yeah. You have you come up with the three minds. That's the most efficient way to communicate what you're trying to communicate. That's three parts. I I just find that so interesting. It always it's it's just a recurring theme in communicating features of reality. Where I see that in terms of the blanket is first there's agent, environment, and blanket. That's the three that's the threesome or the three-fold partition. Well, first there's there's this and that. That's the twofold. Agent, environment just with the arrows connecting them. Then a blanket is interpolated. And then the blanket is nuanced into incoming and outgoing. And so the the the two, the three and the four all represent this sort of canonical um and and also those two three and four are part of a conceptual unity which is the unity of that scenario. So it's like 1 2 3 4 as one two three and four and like that I mean it it it can go so many other ways with the with the numbers in the geometry. Yes, but the low integer numbers are awesome. Yeah, I I of course I have my own epistemology used with this tripart uh distillation that taken as a whole makes a fourth which is echoed in what you just showed on the screen. It's just but what we always have to remember is this isn't a natural law of the world of the universe all of these you know low ordered uh descriptions uh numerical descriptions it's not a natural law of the universe it's a natural law of the way our minds work well and possibly the way generative models work. So exactly well our mind is so you know so we we we'll know best about our own minds let's say but but um but then we can we can maybe see it in other things well we're going to create these generative models according to our conceptualizations right they will work that way too yeah and then if if they're useful in modeling different things as sometimes they do seem to be then that kind of gives them that kind of gives some extension to you know, but but I think maybe to go beyond what Daniel was saying, uh, like with this three minds approach, I think Daniel, I think like you're you're explaining like how it looks for the second mind. And so like

that would be like uh like the zeroth mind would be in terms of needs and responding to needs and you can see things like Masel's hierarchy etc. And then but then the first mind although it's well it it looks at things intellectually surprisingly because that's what it takes to get out of itself like it kind of to let go of itself it becomes intellectual so to speak but then when you have the generative model and you have these two minds in like with action perception then it seems very emotional because that's what that's what the cognition that's what the organism reads uh from the first mind the first mind communicates in terms of emotion and the second mind uh responds with cognition that it imposes but I think in terms of the third mind the see the the first mind doesn't see the second mind the second mind doesn't see the third mind but the third mind sees them and the third mind is this willfulness this cognizance to say hey I have to actively relate these things and then there may be like a god or a default that is also actively somehow relating that I might just be an instrument of let's Okay. And then how do I fit with regard to let's say the default or the god or or whatever like you know so that I'm in sync or or am I separate you know that type of question that's not maybe coming up in these areas but you know so what's maybe that's to say like what's the role of a will related with God you know in terms of trying to uh actively uh manage um I don't think that that's maybe comes coming up there it's it's not mentioned in the textbook I don't think but well you can map what you're saying on to what's said in the but it'd be an extension I think it says that there's and so these symmetries and those like maybe just to add like I talked to my friend Thomas he's a theoretical physicist I talked with uh but it becomes about the conjugate transpose so like the Hamiltonian in these and these are like what they call a free fermionic non-interacting system so it's like a particle all by itself like what are the pipe relationships it could have like with its context. But what's curious is that it has a Hamiltonian, which is what you need in order to have a notion of a coherent self that could evolve through time because you've got to if you're going to evolve through time, you've got to be able to maintain what you are. If you're going to go through different dynamical modes, you have to be able to summarize that, which is what the Hamiltonian does. But the Hamiltonian has this thing where the self- conjugate, it's equal to its own self- conjugate. And so the question is, well, what is that? And the idea is that it's the difference between a a self as a being and a self as its context like your personality or everything that you've invested yourself in. You know, you can identify with either with the spirit you you are that you inhabit something or the context you've created for yourself to inhabit and then you can flip back and forth which is what these quantum symmetries do. And if you as I'm studying it, what happens and I'll conclude but is that those two things which

you're equating you can strip them apart and say oh no let's think of them separately me as a being and me as my context and then let me fit them back together but let's say swap them you know kind of swap them around or make one you know there's different ways you can try to put them together but it's about those how those two things so the generative model and the thing it's modeling can they be inverted to say oh no I'm a fact in the generative model of my environment for example. you know this uh as you're as you're speaking and and Daniel as you're pointing out things in the textbook uh and I you know I've seen these things and and had thoughts about them but I'm seeing new things and having new thoughts about them listening to this discussion and and uh watching Daniel's mind at work pointing out different things at the in the textbook and I'm uh uh struck by the reality of the mind being an interaction with other minds uh just as much as uh self you know internally generated and and I think what that interaction because I've never really thought much about that but I think these games with other people that we play we're like uh you know we switch from I to you and other and all and such like who is the being and who's the context, let's say, or vice versa. Exactly. By communicating, we're we're it kind of forces us to play all these games much more actively. Um in reality, we can't play those games unless we're actually in the presence of others. Yeah, it may be very hard to play that, right? Well, it's it's otherwise it's internal. [Music] Uh I think it certainly helps there's there's compositionality internally and there's compositionality externally like when they talk about these kind of rights of practice and passage. It's like you write down the model, make a simulation and and learn from from what happens. Yes. And that's similar to like you you expect that the restaurant's going to be tasty and maybe maybe it's it is exactly on point. Maybe it surprises you. It's like you you want a hot shower. It could be the perfect temp or you could update. It's like so and then um you so there and then I but then uh the quote at the very beginning of 10 which is that we're we're least aware in general we are least aware of what our minds do best and it's like it is that or how is that true? Why would that be true? Why would awareness, you know, what is awareness and self-awareness such that it is like it's like the joke of the party? It's the least capable, but it's the aware one within an internal compositionality and then transposing that into the extended cognitive setting that you mentioned, Stephen. And it's like we're least aware of what what others are doing. We don't know what others are thinking. Maybe for the best. And and and certainly just looking at um and looking at a a tissue or looking at an ant colony, it's not that each cell or nestmate is solving the problem that the tissue is. Huh? You're putting this into that context of situational awareness or or uh not it's situated awareness. It the meta

awareness is the awareness but you it can't be a thing itself. Yeah. Yeah. Like what is metacognition? But metacognition it's like it's like if you developed a software package with all these like runners, handlers, dispatchers, operators, all these kinds of apparatus, but then it's like and then what is it? Yeah. like what's the terminal rubber hits the road inference. And it's like the whole the whole metacognitive apparatus is just sort of waiting for the key in the ignition for something even if it's just a binary signal coming in or even like a static signal coming in. But but the whole but it's not but the gears don't turn unless something is there. Mhm. And then it's like how slash can we introduce novelty within ourself cuz obviously in a social setting amongst bodied minds we can see how novelty could be introduced. Like you could tell me about a place that you went today that that I would have otherwise not have known of. Okay. But then there's also the compositionality revealed through the novelty of just like I don't know what word I would think of if I were to think what word should I randomly think of. It's like coffee cup or something, you know, again, oh that's cliché or whatever, but it's like but I didn't I didn't know and there's some latent distribution. It's not that I'm going to come up with a word from a language that I've never known. So it's not a good um random number generator for your cryptography per se, but there is some distribution it's sampled from, but what's the status of that of that fancy? Okay, real quick. This is a uh I want to um ascribe a term to what you just portrayed. trying to think of something that you don't know that there there's a principle of something. I don't know the term to describe that. Um uh this is some somehow it's related the end goal of awareness, non-awareness, non-duality. The the the the goal, the the the process of meditation and escaping self is to get to this point of not trying but being of knowing what you don't know. just being awareness which is the state one would have to be in to be able to think of something one doesn't know. I mean that does seem to describe what I've been trying to talk about with the third mind to say like if you let go of all the answers you know like everything you know which would be all your prejudices and if I let go let's say of all of my questions everything I I don't know all these preconceptions this whole generative model and then I just what am I left with? Well, I'm left in this void, this abstract void of how things like that get connected, you know, like what are the possible empty chambers and the doors and those chambers and things like that like where so I'm in this kind of very abstract void that I can get to know and just just be disconnected I guess from everything and just be but be in full cognizance let's say um but maybe so that's that's the state I'm encouraging with an invest stigmatic community. But but maybe that's the state that's the state that Buddhism encourages. I'm not uh sure. It must be if it's if there's a

long, you know, generational millennial tradition of keeping what's true and letting everything else go. It must be the same. And the advantage of that state is that then you can rebuild everything, you know, like just kind of like with a new with a new sense of where is this coming from? Like either you accept it or you or you've kind of reworked it or whatever. And then you can have this harmony between the generative model, the second mind and the first mind reality so to speak or unconscious or or whatever. Well, all of all my takeaway from all of my interaction with you know variational free energy and the active inference the principle of active inference. It's my takeaway is this is a general framework that's universally applicable and it can be applied appropriately or inappropriately depending on how it's put together what the model how the model is constructed. But it that it its power is in its generality and its disadvantage is its generality. Uh and I think it's Go ahead. I think it's begging the idea that it so beautifully gives us freedom. It models the freedom like do you want to perceive? do you want to uh uh act well we'll maximize you know I guess we'll minimize the free energy whatever and then but the idea is that yeah but there could be different solutions that are equally basically you know equal or equal within a very small you know error and so it begs the opportunity that there's a there's somebody who gets to choose you know or somebody who gets to say hey I'm going to make this ambiguous and then given the ambiguity then I'll choose you know what do I which way I'm going to go so because but it but it supplies is a it's a very nice and rare model of freedom uh or the or the preconditions for freedom. Well, my again my conceptualization here is that if it depends on the application the appropriateness. So if it if it's being uh applied to a natural system that's pretty straightforward in terms of physics and and physical constraints. Yeah. It doesn't require free but yeah but if it's being applied to uh um a uh mind state then it has then it's a it's maybe harder to model always there has to be a behavior component I mean that's inherent in this free energy principle right there there has to be an action. You know, if the action is just a concept, then it's more difficult um to construct the model properly. But if the action is interaction with the world, then it's a little bit easier because we're talking about a physical system which is we're going to say a bounded system. Uh may maybe I'll give my just concluding thought that for many systems it does a wonderful job of like uniting action and perception and basically saying like things are going to be a certain way because of how they evolve. And so that's very physical and and maybe even deterministic, but it opens up the idea that uh you can game this whole relationship where like you can balance things out between let's say action or perception or different opportunities and then there could be something that

just if it's able to alter the context, you know, to say, okay, I'm going to reinterpret the context and that'll force things to go one way as opposed another way. like it does not it it sets things up at a tipping point where it's easy to tip you know so it's easy to model freedom in that way it can't you can do that you can do that make that application yes and and that's why I say it could be it can be misleading if it's not constructed properly well so something to code yeah so many great pieces is so many more pieces in linear time that that I know we could add and discuss. H how can something mislead which does not claim to lead? But then I mean e every single little reconfiguration is there when when we go to that sort of trilmma tetral unity duality 1 2 3 4 1 2 3 4 mode on viewing active inference and uh again there's there's more pieces um but a good closure for the cohort. So, thanks fellows for sticking out to the um epistemically delightful culmination and and we'll discuss more in the coming weeks where we go from here. Thank you Dave for leading us on this uh journey in active inference. Indeed. Peace. by